Acids and Bases

Acids and bases are specific types of solutions that have characteristic properties. Initially, they were defined qualitatively, based upon observable properties and reactions. Later the Swedish chemist, **Arrehenius** conceptually defined acids and bases based on the ions they produce in water.

Empirical or Operational (Working) Definitions:

Properties	**Acid**	**Bases**
General	corrosive, conduct electricity	corrosive, conduct electricity, feel slippery
Taste	sour	bitter
reaction with litmus	red stays red; blue turns red	blue stays blue, red turns blue
reaction with bromthymol	yellow	blue
reaction with phenolphthalein	colourless	pink

Chemical Reactivity (MUST KNOW 1 & 4, other two for interest in SCH3U):

1	Acids react with active metals to produce hydrogen and an ionic compound $Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$ $\mathbf{Zn_{(s)} + 2H^+_{(aq)} \rightarrow Zn^{2+}_{(aq)} + H_{2(g)}}$
2	Acids react with carbonates/bicarbonates to produce CO_2, H_2O, ionic compound $CaCO_{3(s)} + 2HNO_{3(aq)} \rightarrow Ca(NO_3)_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$ $\mathbf{CaCO_{3(s)} + 2H^+_{(aq)} \rightarrow Ca^{2+}_{(aq)} + CO_{2(g)} + H_2O_{(l)}}$ $NaHCO_{3(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + CO_{2(g)} + H_2O_{(l)}$ $\mathbf{H^+_{(aq)} + CO_3^-{}_{(aq)} + H^+_{(aq)} \rightarrow CO_{2(g)} + H_2O_{(l)}}$
3	Bases react with ammonium salts to produce $NH_{3(g)}$ $NaOH_{(aq)} + NH_4Cl_{(aq)} \rightarrow NH_{3(g)} + NaCl_{(aq)} + HOH_{(l)}$ $\mathbf{OH^-_{(aq)} + NH_4^+ \rightarrow NH_{3(g)} + HOH_{(l)}}$
4	Acids neutralize bases to make an ionic salt and water $H_2SO_{4(aq)} + 2KOH_{(aq)} \rightarrow H_2O_{(l)} + K_2SO_{4(aq)}$ $\mathbf{2H^+ + OH^-_{(aq)} \rightarrow H_2O_{(l)}}$ $2HC_2H_3O_{2(aq)} + Mg(OH)_{2(s)} \rightarrow \mathbf{Mg(C_2H_3O_2)_{2(aq)} + 2H_2O_{(l)}}$ $\mathbf{2H^+_{(aq)} + Mg(OH)_{2(s)} \rightarrow Mg^{2+}_{(aq)} + 2H_2O_{(l)}}$

1. Theoretical Definitions

Arrhenius Definitions (based on these substances as electrolytes)

Acid	Is a substance that produces hydrogen ions (hydronium ions) in water solution (aqueous) $HCl_{(aq)} \rightarrow H+_{(aq)} + Cl\text{-}(aq)$ $HCl_{(aq)} + H_2O_{(l)} \rightarrow H_3O^+_{(aq)} + Cl^-_{(aq)}$
Base	Is a substance that produces hydroxyl ions (hydroxide ions) in a water solution $NaOH_{(aq)} \rightarrow Na^+_{(aq)} + OH^-_{(aq)}$

The pH Scale

Based upon the operational and theoretical definitions, acids and bases were difficult to quantify. This is because the degree of acidity/basicity depends upon 3 main factors:

1. Strength: degree of ionization producing H^+ or OH^- ions in solution
2. concentration: a strong acid or base's effect can be lessened by diluting the acid with water
3. solubility: a strong acid/bases may have a high degree of ionization, but if it does not dissolve well only a small number of these ions will be in the solution

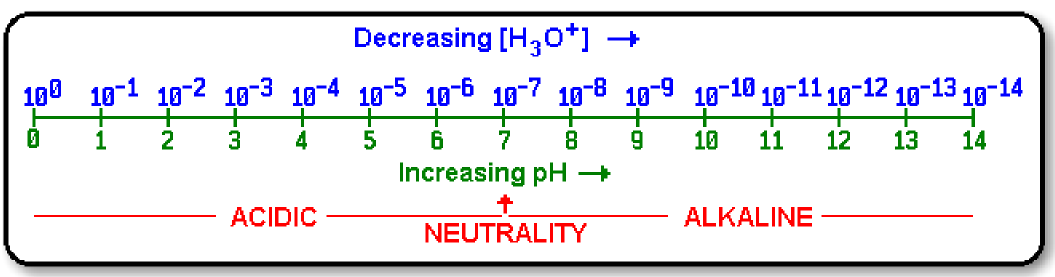
- The pH scale measures the concentration (in moles per litre) of H^+ ions in solution
- In pure water, a very small amount of water molecules self-ionize to break down into equal amounts H^+ or OH^- ions.
- In pure water, the concentration of H^+ ions is 1.0×10^{-7} mol/L
- Due to the huge range of $[H^+_{(aq)}]$ (10^{-14} mol/L to 1.0 mol/L) possible, the pH scale is a logarithmic. This means that **one step** on the pH scale involves a **10x change** in the concentration of H^+ ions.

$$[H^+_{(aq)}] = 10^{-pH}$$

$$pH = -\log_{10}[H^+_{(aq)}]$$

Solution	**$[H^+_{(aq)}]$**	**pH**
Acidic	$>1.0 \times 10^{-7}$ mol/L	< 7
Neutral	$= 1.0 \times 10^{-7}$ mol/L	7
Basic	< is 1.0×10^{-7} mol/L	>7

Note: Be sure you know how to use your calculator (try a youtube for your model) to determine these values.



<u>Testing pH</u>

One convenient method used to measure pH is through the use of indicators. Many substances, including litmus, the one dye almost everyone associates with acids and bases, change colour in response to acid or base. The pigment in red cabbage is another natural substance very commonly used to show colour change. Phenolphthalein is one of the most common indicators used for beginning chemistry, because its colour change is very obvious. There are many other indicators that change colours at different pH's, and so are useful for different purposes. pH paper commonly contains a mixture of different indicators that change colours at different pH's.